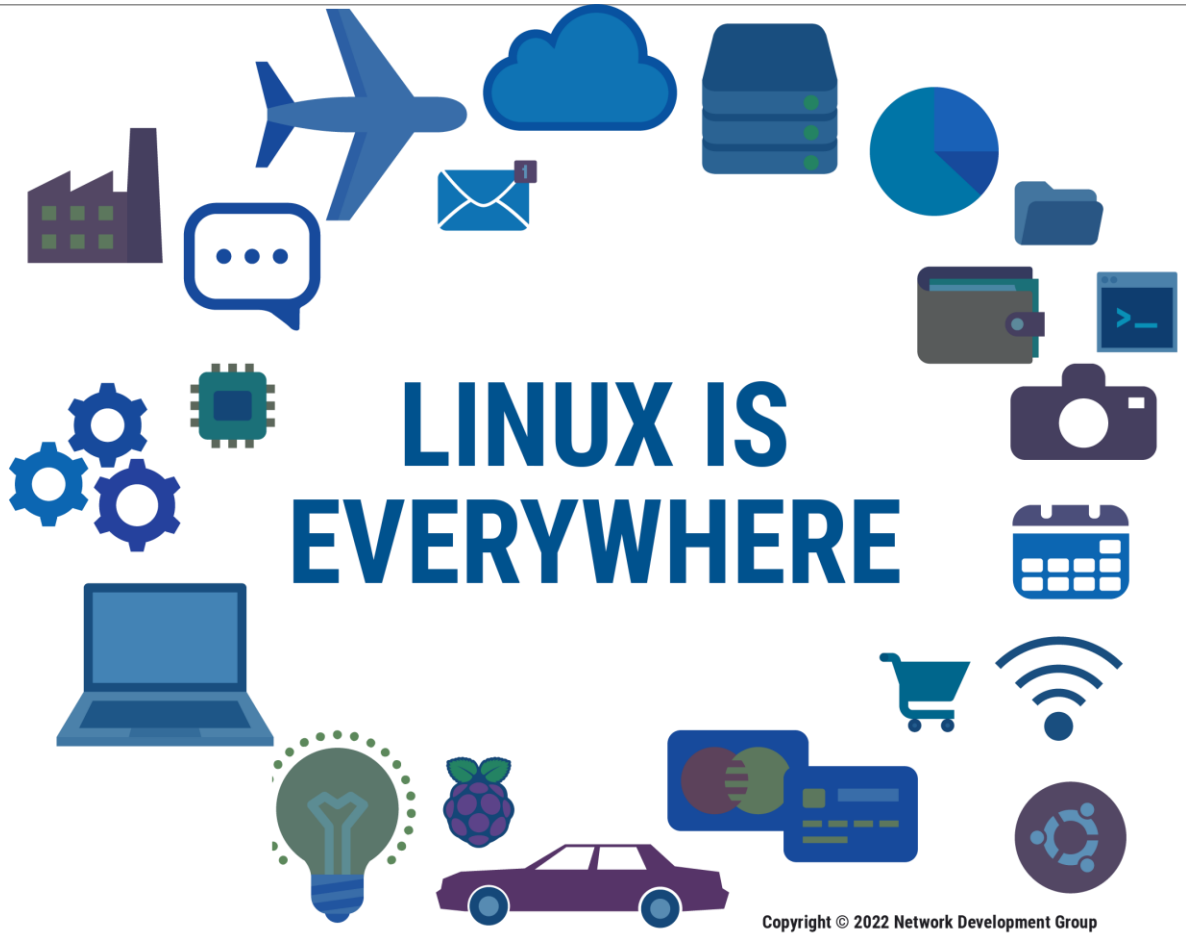
Operating Systems

LINUX DISTRIBUTION

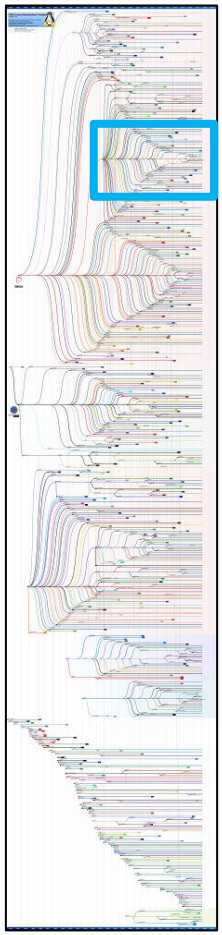
Linux Distribution - Distro

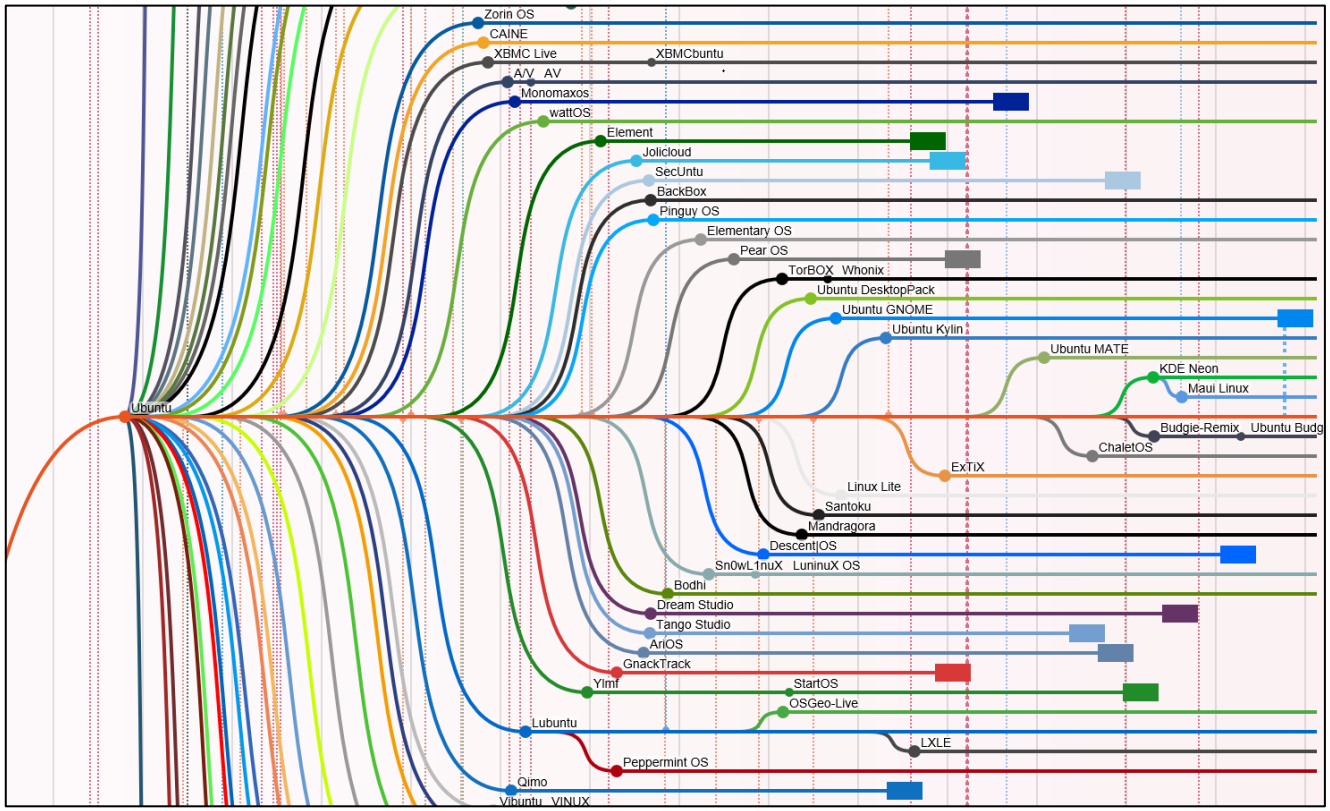


Linux Distribution - Distro

- A Linux distribution (distro) is an operating system made from a software collection, which is based upon the **Linux kernel** and, often, a **package management system**.
- A typical Linux distribution comprises:
  - A Linux kernel
  - GNU tools
  - Others: libraries, additional software, documentation, a window system (the most common being the X Window System), a window manager, and a desktop environment.
- ~ 500/600 distros are in active development.

Timeline of Distro [link]





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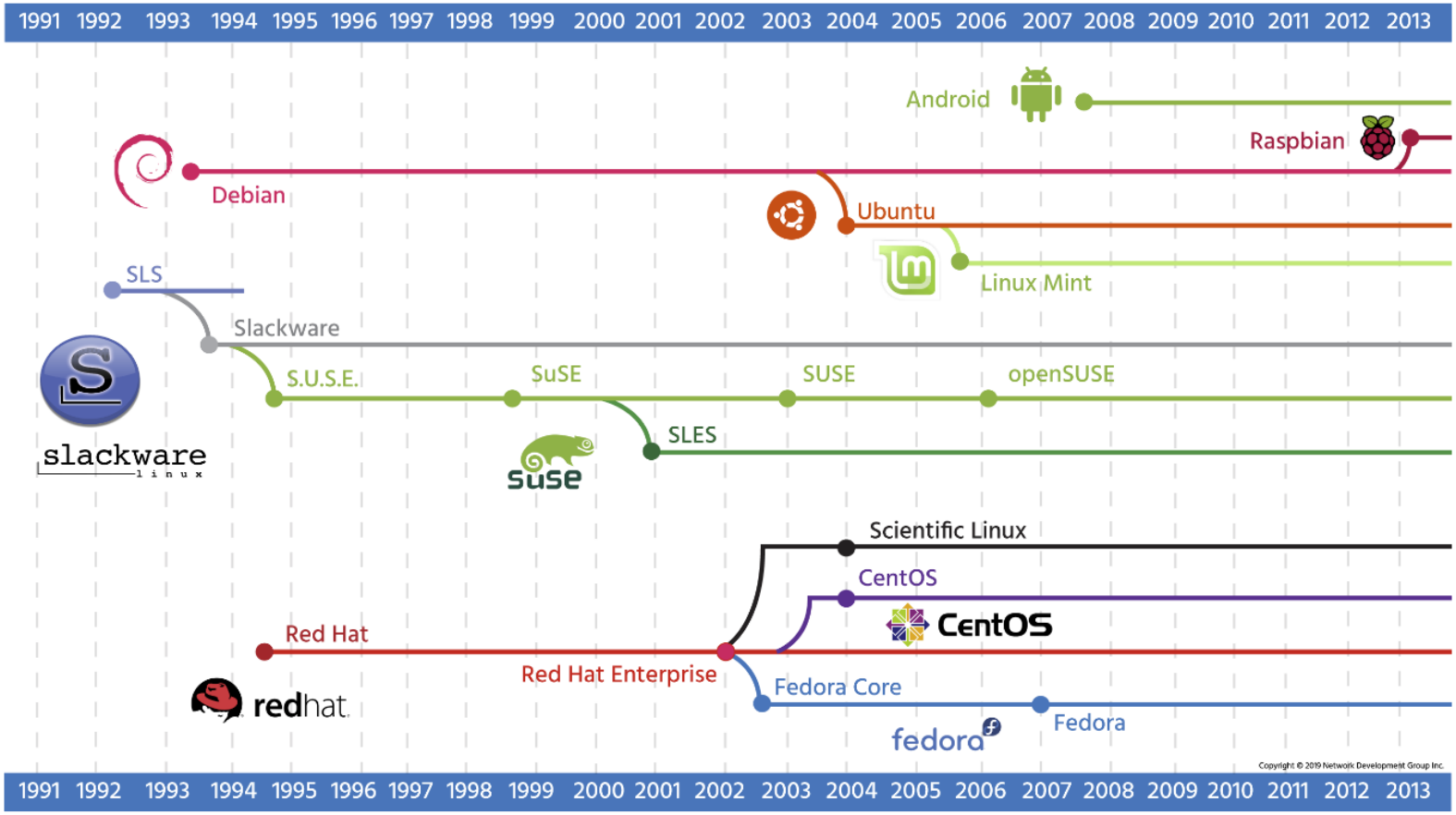
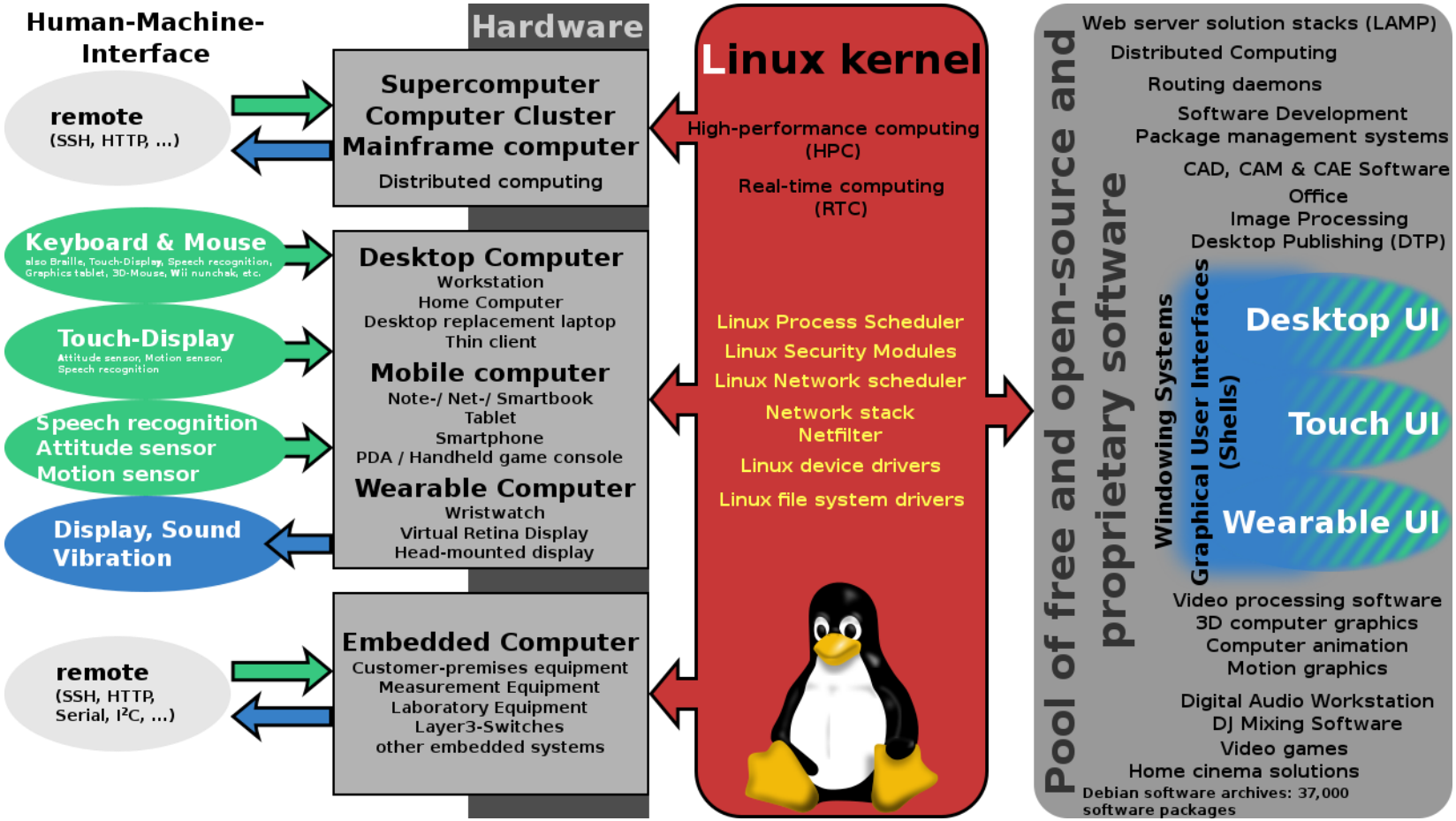


Layers of Linux Distribution

User mode	**User applications**	For example, bash, LibreOffice, GIMP, Blender, 0 A.D., Mozilla Firefox, etc.				
	Low-level system components:	**System daemons**: *systemd, runit, logind, networkd, PulseAudio, ...*	**Windowing system**: *X11, Wayland, SurfaceFlinger (Android)*	**Other libraries:** *GTK+, Qt, EFL, SDL, SFML, FLTK, GNUstep*, etc.		**Graphics**: *Mesa, AMD Catalyst, ...*
	C standard library	`open()`, `exec()`, `sbrk()`, `socket()`, `fopen()`, `calloc()`, ... (up to 2000 subroutines) *glibc* aims to be POSIX/SUS-compatible, *uClibc* targets embedded systems, *bionic* written for Android, etc.				
Kernel mode	**Linux kernel**	`stat`, `splice`, `dup`, `read`, `open`, `ioctl`, `write`, `mmap`, `close`, `exit`, etc. (about 380 system calls) The Linux kernel System Call Interface (SCI, aims to be POSIX/SUS-compatible)				
		Process scheduling subsystem	IPC subsystem	Memory management subsystem	Virtual files subsystem	Network subsystem
		Other components: ALSA, DRI, evdev, LVM, device mapper, Linux Network Scheduler, Netfilter Linux Security Modules: *SELinux*, *TOMOYO*, *AppArmor*, *Smack*				
Hardware (CPU, main memory, data storage devices, etc.)						

Linux Kernel



Linux Origin

Linus Torvalds
1969 - Present

- **Linus Torvalds** developed 1st version 0.01 in 1991.
- Linux was initially distributed as source code only.
- Later as a pair of downloadable floppy disk images:
  - A bootable and containing the Linux kernel
  - The other with a set of GNU utilities and tools for setting up a file system.

Package Management

- Distributions are normally segmented into packages. Each package contains a specific application or service.
- **Packages** are bundles of software and metadata such as the software's full name, description of its purpose, version number, checksum, and a list of dependencies necessary for the software to run properly
- **A Package management system (package manager)** is a collection of software tools that automate the process of installing, upgrading, configuring, and removing computer programs for a computer's operating system in a consistent manner

DPKG

- **DPKG** is a package manager for Debian-based systems. It can install, remove, and build packages, but unlike other package management systems, it cannot automatically download and install packages or their dependencies.
- Detail of DPKG usage: `man dpkg`
- Example of DPKG:
  - `dpkg -l` list all installed packages
  - `dpkg -l | grep xxx` check if a specific xxx is installed
  - `sudo dpkg -i xxx.deb` install xxx.deb
  - `sudo dpkg -r xxx` uninstall xxx

APT

- The **`apt`** command is a powerful command-line tool, which works with Ubuntu's Advanced Packaging Tool (APT) performing such functions as installation of new software packages, upgrade of existing software packages, updating of the package list index, and even upgrading the entire Ubuntu system **remotely**.
- Detail of APT usage: `apt help`
- Example of DPKG:
  - `sudo apt install xxx` install xxx
  - `sudo apt remove xxx` remove xxx
  - `sudo apt update` update the local package index
  - `sudo apt upgrade` update packages and system

Shell

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Why is knowing the command line important?

By understanding the foundation of Linux, you have the ability to work on ANY Linux distribution. This could mean one company with a mixed environment or a new company with a different Linux distribution.

What is a shell?

1. Once a user has entered a command the terminal then accepts what the user has typed and passes it to a *shell*.
2. The shell is the command line interpreter that translates commands entered by a user into actions to be performed by the operating system.

What is a shell?

1. When a terminal application is run, and a shell appears, displaying an important part of the interface—the *prompt.*
2. The structure of the prompt may vary between distributions, but typically contains information about the user and the system. Below is a common prompt structure:

```
sysadmin@localhost:~$
```

What is a shell?

User Name:

```
sysadmin@localhost:~$
```

System Name:

```
sysadmin@localhost:~$
```

Current Directory:

```
sysadmin@localhost:~$
```

The ~ symbol is used to as shorthand for the user's home directory.